

SteelVision AI

Explainable Computer Vision for Automated Steel Surface Inspection

VALIDATION ACCURACY

98.61%

MACRO F1 SCORE

98.60%

DEFECT CLASSES

6

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Why This Matters

Current Challenges

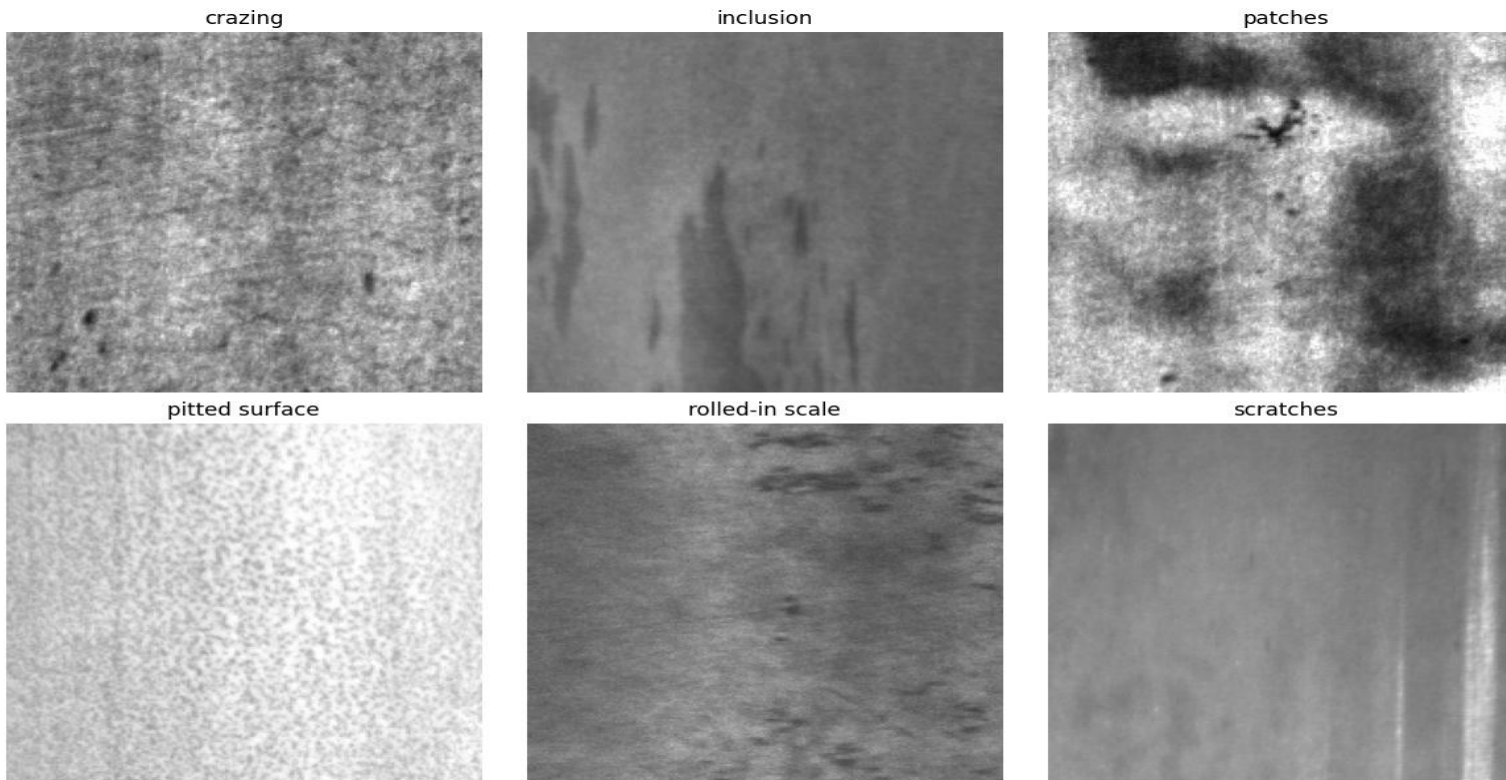
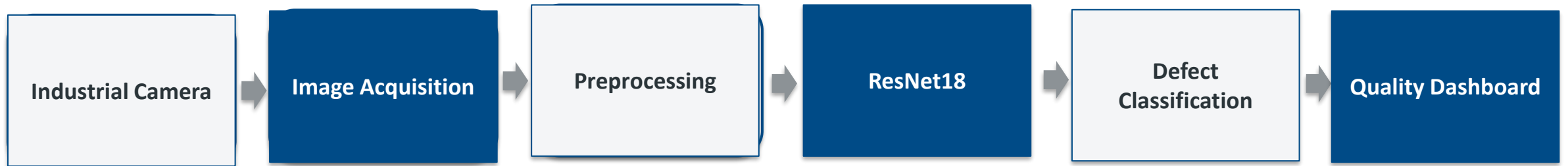
- Manual inspection forms a critical production bottleneck.
- Human fatigue leads to highly inconsistent QA.
- Subjective quality assessments vary between shifts.
- Manual processes cannot scale with modern line speeds.

Business Impact

- Reduced manual inspection effort
- Faster identification of surface anomalies
- Improved product quality consistency
- Foundation for Industry 4.0 quality systems



The AI Solution Architecture



Dataset: NEU Surface Defect Dataset

1800 Images

1440 Training

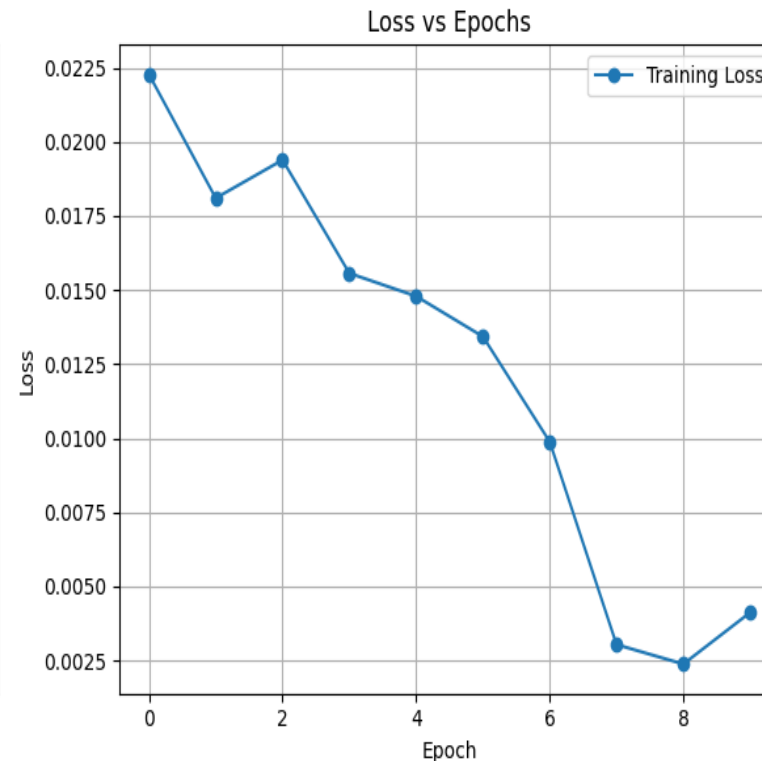
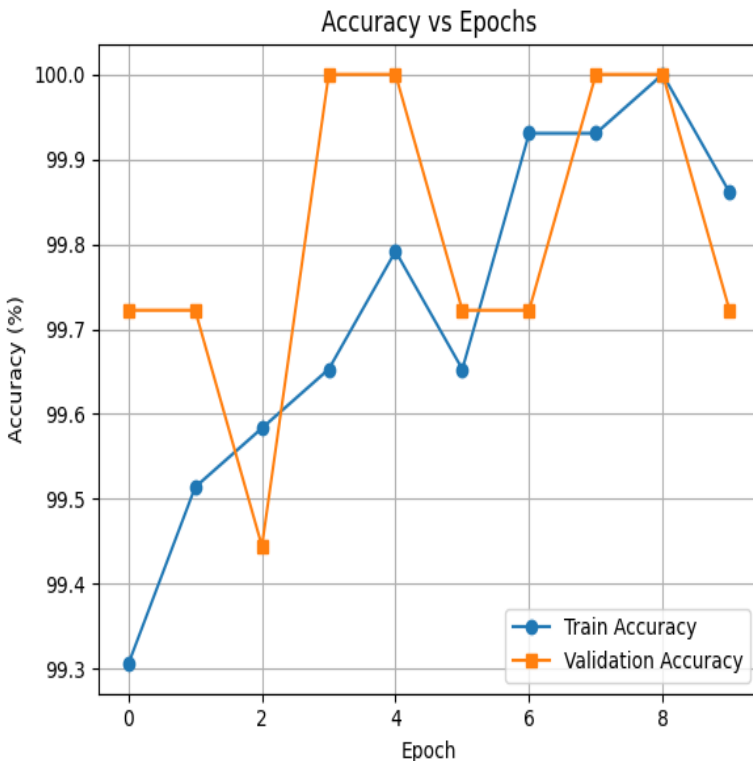
360 Validation

6 Defect Classes

Model Performance

ACCURACY
98.61%

MACRO F1
98.60%



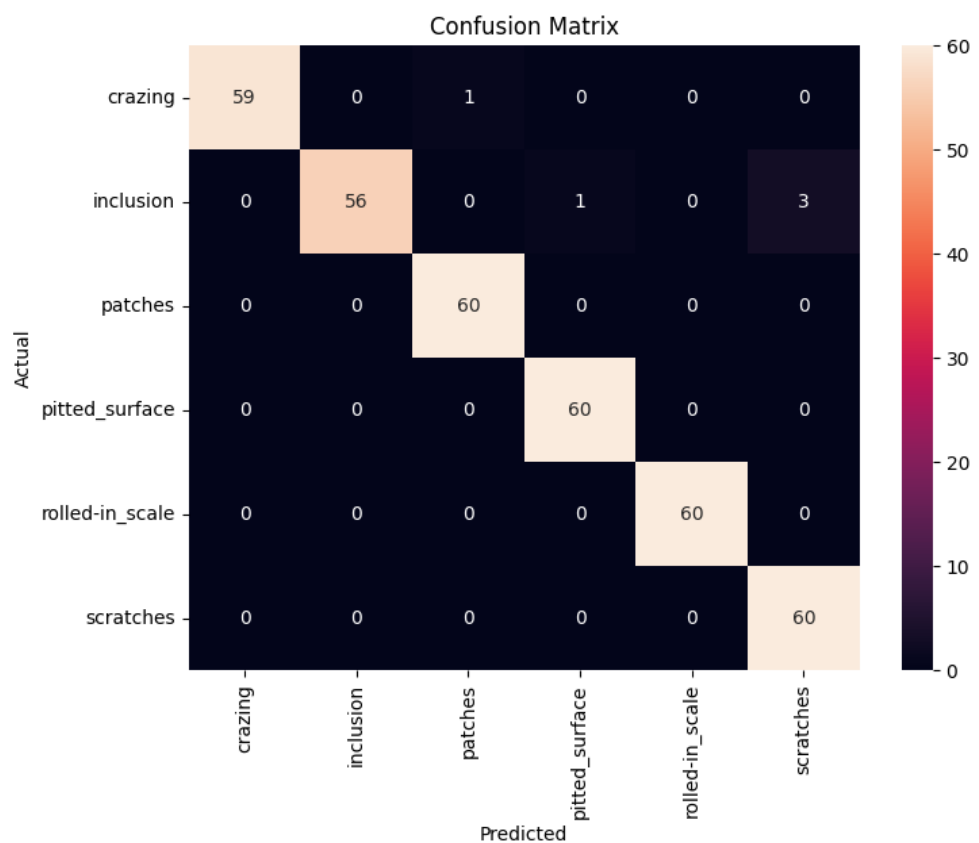
Performance Highlights

- Stable convergence across epochs confirms optimal learning rate.
- Near-perfect classification on unseen validation data.
- Strong validation consistency indicates minimal overfitting.
- Robust multi-class performance despite morphological similarities.

Key Findings

- **Rapid convergence**
- **Stable learning**
- **Minimal overfitting**
- **Strong generalization**

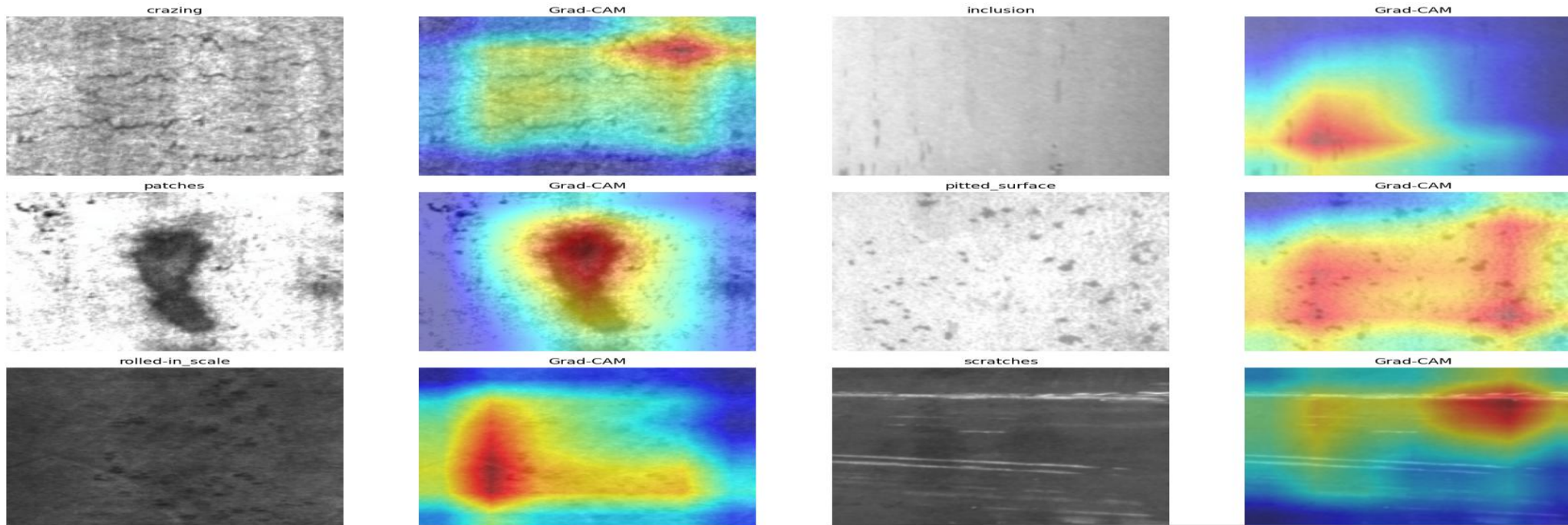
Classification Results



- Executive Insights**
- Strong diagonal class separation.
 - Minimal inter-class confusion.
 - High reliability across all 6 defect categories.
 - No major false-positive clusters detected.

...	precision	recall	f1-score	support
crazing	1.00	0.98	0.99	60
inclusion	1.00	0.93	0.97	60
patches	0.98	1.00	0.99	60
pitted_surface	0.98	1.00	0.99	60
rolled-in_scale	1.00	1.00	1.00	60
scratches	0.95	1.00	0.98	60
accuracy			0.99	360
macro avg	0.99	0.99	0.99	360
weighted avg	0.99	0.99	0.99	360

Explainable AI (XAI)



Key Insight

"Grad-CAM analysis confirms that the model focuses on physically meaningful defect regions rather than background texture."

Crazing → Crack Structures

Inclusion → Localized Inclusions

Patches → Intensity Variations

Pitted Surface → Distributed Pits

Rolled-In Scale → Scale Patterns

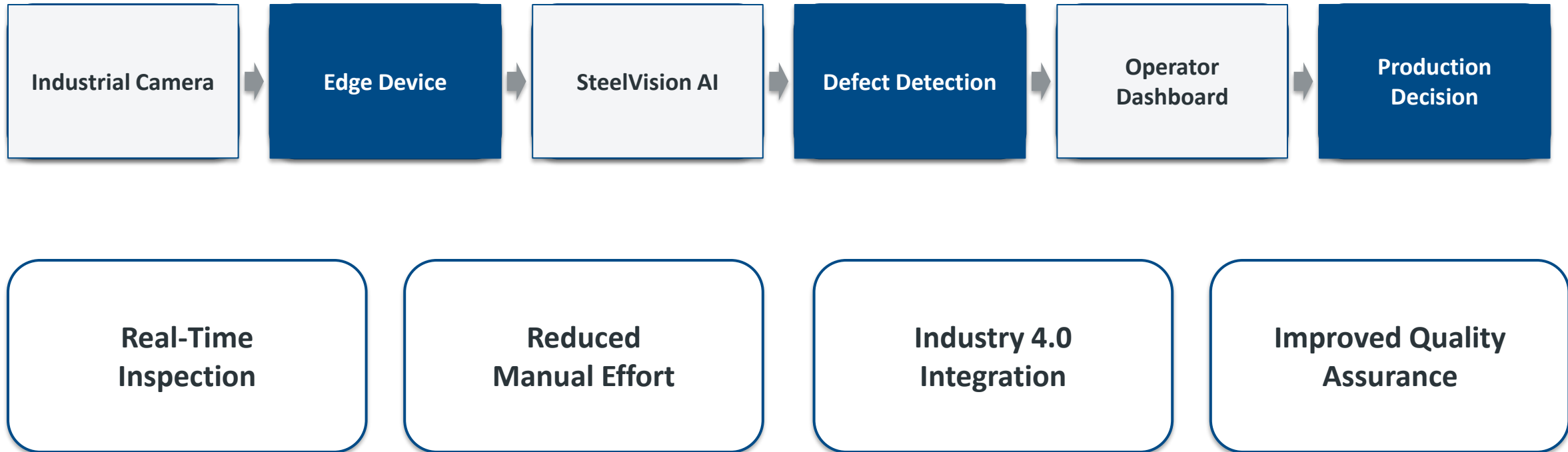
Scratches → Linear Defect Geometry

✓ Transparent
Decision-
Making

✓ Defect-
Focused
Attention

✓ Deployment-
Ready
Explainability

Potential Tata Steel Deployment



This project demonstrates the feasibility of deploying explainable computer vision systems for automated steel surface inspection in modern manufacturing environments.